🔥 **FINAL TOP 10 QUESTIONS (COMBINED MODULE 1 + MODULE 3)**

✅ **1. Explain the concept of Artificial Intelligence with suitable examples.**

👉 (Module 1 – CO2)

✅ **2. Describe how Artificial Intelligence works in real-world applications.**

👉 (Module 1 – CO2)

✅ **3. Differentiate between Narrow AI and General AI.**

👉 (Module 1 – CO2) ⚠️ VERY IMPORTANT THEORY

✅ **4. Analyze the different types of Agents in AI and its environment. Identify percepts, actions and goals with an example.**

👉 (Module 1 – CO4) ⭐ HIGH LEVEL / HIGH WEIGHT

✅ **5. Apply Best First Search algorithm to solve a given graph problem and trace the steps.**

👉 (Module 1 – CO3) ⭐ ALGORITHM (VERY LIKELY)

✅ **6. *Solve a pathfinding problem using A algorithm and explain how f(n) is calculated.****

👉 (Module 1 – CO3) 🔥 MUST STUDY

✅ **7. Explain the need for Machine Learning in solving real-world problems.**

👉 (Module 3 – CO2)

✅ **8. Explain the relationship between Machine Learning and Artificial Intelligence.**

👉 (Module 3 – CO2) ⚠️ FREQUENT QUESTION

✅ **9. Explain the differences between structured, semi-structured, and unstructured data with examples.**

👉 (Module 3 – CO2) ⭐ EASY FULL MARKS

✅ **10. Apply the concept of data visualization to represent sales trends using Bar chart, Pie chart, Histogram, and Dot Plot.**

👉 (Module 3 – CO3) 🔥 CONFIRMED TYPE

🔥 **BONUS (VERY HIGH CHANCE – STUDY IF TIME)**

👉 If they twist paper, these can replace above:

- Analyze evolution of AI (CO4)
- Machine Learning process steps
- IQR / Five-point summary (numerical)
- Descriptive vs Predictive vs Prescriptive analytics

🔥 **Q1. Explain the concept of Artificial Intelligence with suitable examples. (8 Marks)**

Answer:

Artificial Intelligence (AI) is a branch of computer science that focuses on designing intelligent machines that can simulate human cognitive abilities such as learning, reasoning, problem-solving, perception, and decision-making.

According to John McCarthy, AI is defined as *"the science and engineering of making intelligent machines."*

Core Characteristics of AI:

1. **Learning:** Ability to acquire knowledge from data (Machine Learning)
2. **Reasoning:** Logical decision-making capability
3. **Problem Solving:** Finding optimal solutions
4. **Perception:** Interpreting sensory inputs (vision, speech)
5. **Self-correction:** Improving performance using feedback

How AI Works:

AI systems use:

- Large datasets
- Algorithms (ML, deep learning)
- Computational power

They identify patterns in data and make predictions or decisions.

Examples:

- **Virtual Assistants:** Siri, Alexa
- **Self-driving Cars:** Detect objects and navigate
- **Recommendation Systems:** Netflix, Amazon
- **Fraud Detection:** Banking systems

Applications:

- Healthcare (diagnosis)
- Finance (risk analysis)
- Robotics
- Natural Language Processing

Conclusion:

AI enables machines to mimic human intelligence and is transforming industries by improving automation and efficiency.

🔥 **Q2. Describe how Artificial Intelligence works in real-world applications. (8 Marks)**

Answer:

Artificial Intelligence works by integrating **data, algorithms, and computational models** to simulate intelligent behavior.

Working Process:

1. **Data Collection:**
   Data is collected from various sources (sensors, databases, internet)
2. **Data Preprocessing:**
   Cleaning, normalization, and transformation of data
3. **Training Phase:**
   Algorithms learn patterns using training data
4. **Model Building:**
   A predictive model is created
5. **Decision Making:**
   Model outputs predictions or decisions
6. **Feedback and Improvement:**
   Model improves through continuous learning

Key Components:

- Machine Learning
- Neural Networks
- Natural Language Processing

Real-world Applications:

- **Healthcare:** Disease prediction using patient data
- **E-commerce:** Recommendation systems
- **Transportation:** Autonomous vehicles
- **Finance:** Fraud detection

Conclusion:

AI systems continuously learn from data and improve performance, making them suitable for real-world problem-solving.

🔥 **Q3. Differentiate between Narrow AI and General AI. (8 Marks)**

Answer:

Artificial Intelligence can be classified into Narrow AI and General AI based on capability.

Comparison Table:

Feature	Narrow AI (Weak AI)	General AI (Strong AI)
Definition	Designed for specific tasks	Performs any intellectual task like humans
Scope	Limited	Broad
Intelligence	Task-specific	Human-like intelligence
Learning	Pre-trained for tasks	Learns and adapts universally
Flexibility	Low	High
Examples	Alexa, Google Maps	Not yet developed

Feature	Narrow AI (Weak AI)	General AI (Strong AI)
Status	Exists today	Theoretical

Explanation:

- Narrow AI is specialized and efficient in a single domain
- General AI aims to replicate full human intelligence

Conclusion:

Current AI systems are Narrow AI, while General AI remains a future goal.

🔥 **Q4. Analyze the different types of Agents and Environment with example. (8 Marks)**

Answer:

An **agent** is an entity that perceives its environment through sensors and acts upon it using actuators to achieve a goal.

Agent Components:

- **Percepts:** Inputs from environment
- **Actions:** Outputs

Types of Agents:

1. **Simple Reflex Agent:**
   Acts based on current percept
2. **Model-Based Agent:**
   Maintains internal state
3. **Goal-Based Agent:**
   Acts to achieve goals
4. **Utility-Based Agent:**
   Chooses best action based on utility

Environment Types:

- Fully observable / Partially observable
- Deterministic / Stochastic
- Static / Dynamic

Example: Self-driving Car

- Percepts → Sensors
- Actions → Steering, braking
- Goal → Safe travel

Analysis:

Different agent types handle environments with increasing complexity and intelligence.

Conclusion:

Agents interact intelligently with environments to achieve optimal outcomes.

🔥 **Q5. Apply Best First Search Algorithm. (8 Marks) (LAB PRG)**

Answer:

Best First Search is an informed search algorithm that selects the most promising node using a heuristic function.

Evaluation Function:

f(n) = h(n)

Algorithm Steps:

1. Start with initial node
2. Add to OPEN list

3. Select node with lowest h(n)
4. Expand node
5. Repeat until goal

Example:

Nodes:
A(10), B(5), C(7)
➞ Expand B first

Analysis:

- Uses heuristic guidance
- Faster than blind search

Advantages:

- Efficient

Disadvantages:

- Not always optimal

Conclusion:

Best First Search improves performance but lacks guaranteed optimality.

🔥 ***Q6. Explain A* Algorithm. (8 Marks)**** **(LAB PRG)**

Answer:

A* algorithm is an informed search algorithm that finds the optimal path by combining actual cost and heuristic.

Formula:

$$f(n) = g(n) + h(n)$$

Steps:

1. Start node
2. Compute f(n)
3. Expand lowest node
4. Repeat

Properties:

- Complete
- Optimal

Applications:

- Navigation systems
- Games

Conclusion:

A* is widely used due to efficiency and optimal results.

🔥 **Q7. Explain the need for Machine Learning. (8 Marks)**

Need for Machine Learning:

Machine Learning is required because traditional programming approaches fail to handle complex, large-scale, and dynamic data.

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1. Limitation of Traditional Programming

In traditional systems, rules must be explicitly programmed.
However, for complex problems like image recognition or speech processing, defining rules manually is not feasible.

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2. Handling Large Volumes of Data

Modern systems generate massive data (Big Data).
ML can process and extract meaningful insights efficiently.

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3. Pattern Recognition

ML identifies hidden patterns in data which are not easily visible using traditional methods.

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4. Prediction and Decision Making

ML models can predict future outcomes such as:

- Stock prices
- Weather forecasting

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5. Automation and Adaptability

ML systems automatically improve with experience without human intervention.

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6. Real-world Applications

- Spam detection (email filtering)
- Recommendation systems (Netflix, Amazon)
- Medical diagnosis

✅ **Q10. Apply the concept of data visualization to show how sales trends can be represented using Bar chart, Pie chart, Histogram, Dot Plot. (8 Marks)**

Answer:

Data visualization is the graphical representation of data to analyze trends, patterns, and insights.

To **apply the concept**, consider the following **monthly sales dataset (in ₹ thousands):**

Month	Sales
Jan	100

Month	Sales
Feb	150
Mar	200
Apr	250
May	300

🔹 **1. Bar Chart (Comparison of Sales Over Time)**

- X-axis → Months
- Y-axis → Sales values
- Each bar represents sales of a month

👉 **Analysis:**

- Clearly shows increasing trend
- Easy comparison between months

👉 **Conclusion:**
Bar chart is best for comparing discrete categories like monthly sales.

🔹 **2. Pie Chart (Proportion of Total Sales)**

First calculate total sales = 100 + 150 + 200 + 250 + 300 = **1000**

Now percentage contribution:

Month	% Contribution
Jan	10%
Feb	15%
Mar	20%
Apr	25%
May	30%

👉 **Analysis:**

- May contributes highest share
- Jan contributes least

👉 **Conclusion:**
Pie chart shows **relative contribution of each category**.

🔹 **3. Histogram (Distribution of Sales Values)**

Group data into intervals (bins):

Range	Frequency
100–150	2
150–250	2
250–350	1

👉 **Analysis:**

- Most values lie between 100–250
- Shows spread and concentration

👉 **Conclusion:**
Histogram is used to understand **data distribution**.

🔹 **4. Dot Plot (Individual Data Representation)**

Data points:
100, 150, 200, 250, 300

👉 Representation:

- Each value plotted as a dot on a number line

👉 **Analysis:**

- Clearly shows increasing pattern
- No overlap in values

👉 **Conclusion:**
Dot plot is useful for **small datasets and identifying patterns quickly**.

🔹 **Overall Analysis of Sales Trend:**

- Sales are **steadily increasing** month by month
- Highest growth observed in later months
- No fluctuation → consistent growth pattern

🔹 **Final Conclusion:**

Different visualization techniques provide different insights:

- Bar chart → comparison
- Pie chart → proportion
- Histogram → distribution
- Dot plot → individual values

👉 Hence, data visualization helps in **better decision-making and trend analysis**.

🔥 **Q8. Difference between Artificial Intelligence and Machine Learning (8 Marks)**

Answer:

No.	Artificial Intelligence (AI)	Machine Learning (ML)
1	AI is a broad field of creating intelligent machines	ML is a subset of AI
2	Goal is to simulate human intelligence	Goal is to enable learning from data
3	Includes reasoning, problem-solving, NLP, robotics	Focuses mainly on data-driven learning
4	Can work without learning (rule-based systems)	Always requires learning from data
5	Uses logic, rules, and algorithms	Uses statistical models and algorithms
6	Example: Expert systems, robotics	Example: Recommendation systems
7	Aims to mimic human thinking and behavior	Aims to improve performance through experience
8	Broader scope	Narrower scope

No.	Artificial Intelligence (AI)	Machine Learning (ML)
9	May or may not involve ML	Always a part of AI

🔥 **Q9. Difference between Structured, Semi-Structured, and Unstructured Data (8 Marks)**

Answer:

No.	Structured Data	Semi-Structured Data	Unstructured Data
1	Highly organized format	Partially organized	No predefined structure
2	Stored in tables (rows & columns)	Stored in tags or key-value pairs	Stored in raw formats
3	Easy to store and query	Moderate complexity	Difficult to process
4	Uses relational databases (SQL)	Uses NoSQL databases	Stored in data lakes
5	Schema is fixed	Schema is flexible	No schema
6	Example: Excel sheets, SQL tables	Example: JSON, XML	Example: Images, videos, text
7	Easy to analyze	Requires some processing	Requires advanced techniques
8	High consistency	Medium consistency	Low consistency
9	Supports simple queries	Supports semi-complex queries	Requires ML/NLP for analysis